

5G RAN

PCI Conflict Detection and Self-Optimization Feature Parameter Description

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1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 5G RAN10.1 Draft A (2025-12-31)

This issue does not introduce any changes to 5G RAN9.1 01 (2025-03-10).

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following feature.

Feature ID	Feature Name	Chapter/Section
FOFD-021204	Automatic Neighbour Relation (ANR)	4 PCI Conflict Detection and Self-Optimization

2.3 Differences Between NSA and SA

Function Name	Difference	Chapter/Section
PCI conflict detection and self-optimization	• PCI confusion detection is supported between NR cells or between neighboring LTE cells of an NR cell in SA networking, whereas PCI confusion detection is supported only between NR cells in NSA networking. • PCI conflict detection based on Xn messages is supported only in SA networking and is not supported in NSA networking.	4 PCI Conflict Detection and Self-Optimization

2.4 Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
PCI conflict detection and self-optimization	None	4 PCI Conflict Detection and Self-Optimization

3 Overview

3.1 Background

UEs use physical cell identifiers (PCIs) to distinguish between the radio signals of different cells. Each New Radio (NR) cell has one PCI, and 1008 PCIs are available in an NR system. If a network that serves many NR cells does not have enough PCIs, the PCIs have to be reused, meaning that multiple NR cells that use the same PCI. In this case, certain principles (such as distance requirements) must be adhered to; otherwise, there will be PCI conflicts between intra-frequency NR cells that use the same PCI, leading to UE service drops or handover failures.

To eliminate or reduce PCI conflicts, Huawei has introduced PCI conflict detection and self-optimization. This function contains two sub-functions: PCI conflict detection and PCI self-optimization.

- PCI conflict detection: The gNodeB automatically detects PCI conflicts between NR cells or between neighboring LTE cells of an NR cell.
- PCI self-optimization: The MAE-Access automatically reallocates appropriate PCIs to conflicting NR cells (cells with PCI conflicts) based on the PCI conflict information reported by the gNodeB. For neighboring LTE cells with PCI conflicts of an NR cell, PCIs can be manually but not automatically reallocated.

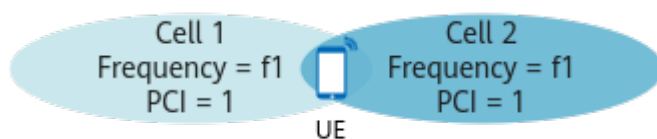
3.2 Basic Concepts

PCI conflicts include PCI collisions and PCI confusions.

PCI Collision

PCI collisions occur when two or more intra-frequency NR cells that are within close proximity to each other use the same PCI. This leads to mutual signal interference in cell overlapping areas where UEs fail to correctly demodulate signals, increasing service drop rates on the live network. As shown in [Figure 3-1](#), a PCI collision occurs between cells 1 and 2 working on the same frequency and using the same PCI.

Figure 3-1 PCI collision

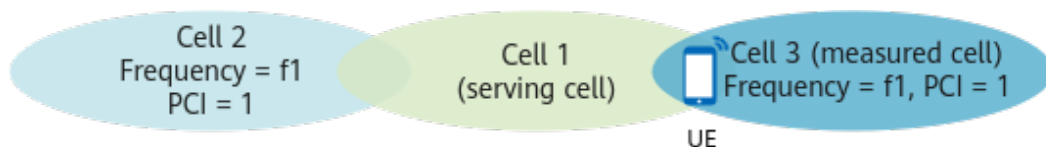


PCI Confusion

When a UE moves across cells, the gNodeB instructs the UE to measure neighboring cells on a specified frequency, and the UE reports the PCI of the cell that meets signal quality requirements to the gNodeB. When the intra-RAT or inter-RAT neighboring relation table (NRT) of the serving cell contains two or more intra-frequency cells that use the same PCI, the gNodeB cannot determine the neighboring cells that the UE has measured based on the reported PCIs and therefore cannot determine the target cell for the UE handover if the UE does not support neighboring cell measurements with the *reportCGI* report type or intra-RAT or inter-RAT automatic neighbor relation (ANR) has not been enabled for the serving cell. In this case, the UE handover will fail. This type of PCI conflict is referred to as PCI confusion.

Figure 3-2 illustrates an example where cell 2 and cell 3 are intra-frequency NR cells or intra-frequency neighboring LTE cells that use the same PCI and they are both contained in the intra-RAT NRT or inter-RAT NRT of cell 1 (the serving NR cell). When the UE moves from cell 1 to cell 3 which is measured and meets the handover conditions, the gNodeB receives a measurement report that contains the PCI of cell 3. However, the NRT of cell 1 shows that this PCI has been used by both cell 2 and cell 3. Therefore, the gNodeB fails to determine whether cell 2 or cell 3 is the target cell to which the UE is handed over and the UE handover cannot be initiated. A PCI confusion occurs between cell 2 and cell 3.

Figure 3-2 PCI confusion



NOTE

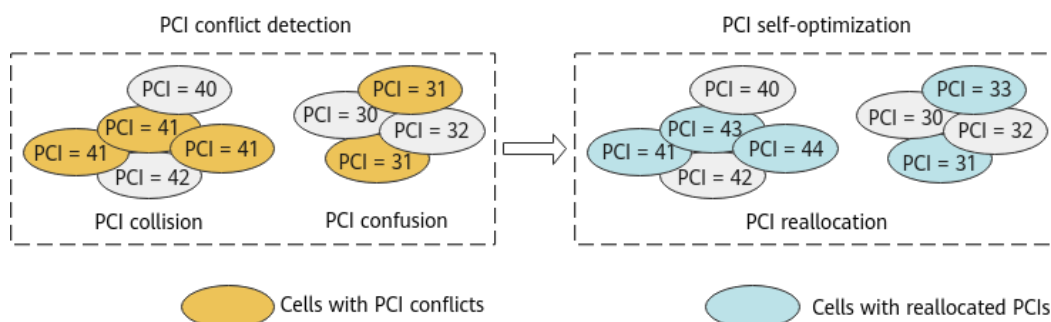
If the UE supports neighboring cell measurements with the *reportCGI* report type and intra-RAT or inter-RAT ANR has been enabled for the serving NR cell, the gNodeB instructs the UE to measure neighboring cells on a specified frequency. After receiving the measurement report containing the PCI of the cell that meets the signal quality requirements (PCI of cell 3) from the UE, the gNodeB instructs the UE to start the measurement with the *reportCGI* report type. Once obtaining the *CGI-Info* of the cell with the specified PCI (cell 3), the UE reports it to the gNodeB. The gNodeB then determines that cell 3 is the target cell for UE handovers and completes the handover procedure. For more details, see *ANR*.

4 PCI Conflict Detection and Self-Optimization

4.1 Principles

PCI conflict detection and self-optimization includes PCI conflict detection and PCI self-optimization, as shown in [Figure 4-1](#).

Figure 4-1 PCI conflict detection and self-optimization



- PCI conflict detection
 - PCI conflicts between NR cells include PCI collisions and PCI confusions. The gNodeB automatically detects the PCIs of NR cells to identify PCI collisions and PCI confusions between NR cells, and reports the identified PCI conflicts to the MAE-Access.
 - PCI conflicts between neighboring LTE cells of an NR cell can only be PCI confusions. The gNodeB automatically detects the PCIs of the neighboring LTE cells to identify PCI confusions between the neighboring LTE cells, and reports the identified PCI conflicts to the MAE-Access.
- PCI self-optimization
 - The MAE-Access automatically reallocates appropriate PCIs to NR cells with PCI conflicts based on the PCI conflict information reported by the gNodeB, and then delivers reallocated PCIs to the gNodeB.

- The MAE-Access cannot automatically reallocate appropriate PCIs to neighboring LTE cells of an NR cell if the neighboring LTE cells have PCI conflicts. In this situation, manual handling is required.

4.1.1 PCI Conflict Detection

PCI conflict detection is controlled by the **NRCellAlgoSwitch.AnrSwitch** parameter.

- If the **PCI_COLLISION_DETECT_SW** option is selected, the gNodeB can detect PCI collisions between NR cells.
- If the **PCI_CONFUSION_DETECT_SW** option is selected, the gNodeB can detect PCI confusions between NR cells.
- If the **EUTRAN_PCI_CONFUSION_DETECT_SW** option is selected, the gNodeB can detect PCI confusions between neighboring LTE cells of an NR cell.

The PCI conflict detection procedure consists of triggering of PCI conflict detection, PCI conflict identification, and PCI conflict reporting.

4.1.1.1 Triggering of PCI Conflict Detection

The gNodeB starts PCI conflict detection when any of the following conditions is met:

- Local cell: A local cell is added or deleted, the PCI of a local cell is changed, or the operating frequency of a local cell is changed.
- NCL: An intra-RAT external cell or external LTE cell is added or deleted, the PCI of an intra-RAT external cell or external LTE cell is changed, or the operating frequency of an intra-RAT external cell or external LTE cell is changed.
- NRT: A neighbor relationship with an intra-RAT cell or LTE cell is added or deleted, the PCI of an intra-RAT neighboring cell or neighboring LTE cell is changed, or the operating frequency of an intra-RAT neighboring cell or neighboring LTE cell is changed.

PCI conflict detection is classified into the following types by triggering method:

- PCI conflict detection based on manual modification
Manual modification of NE parameters results in the preceding changes, triggering PCI conflict detection. For details, see [Triggering of PCI Conflict Detection Based on Manual Modification](#).
- PCI conflict detection based on ANR
The gNodeB uses ANR to automatically maintain external cells and neighbor relationships, resulting in the preceding NCL and NRT changes and in turn triggering PCI conflict detection. For details, see [Triggering of PCI Conflict Detection Based on ANR](#).
- PCI conflict detection based on Xn messages
gNodeBs exchange information through Xn messages. An Xn interface setup or gNodeB configuration update at either end of the Xn interface results in the preceding NCL changes, triggering PCI conflict detection. For details, see [Triggering of PCI Conflict Detection Based on Xn Messages](#).

Triggering of PCI Conflict Detection Based on Manual Modification

If manual modification of NE parameters results in local cell, NCL, or NRT changes associated with PCI conflicts, the gNodeB starts PCI conflict detection.

Triggering of PCI Conflict Detection Based on ANR

PCI conflict detection based on ANR can work either proactively or passively.

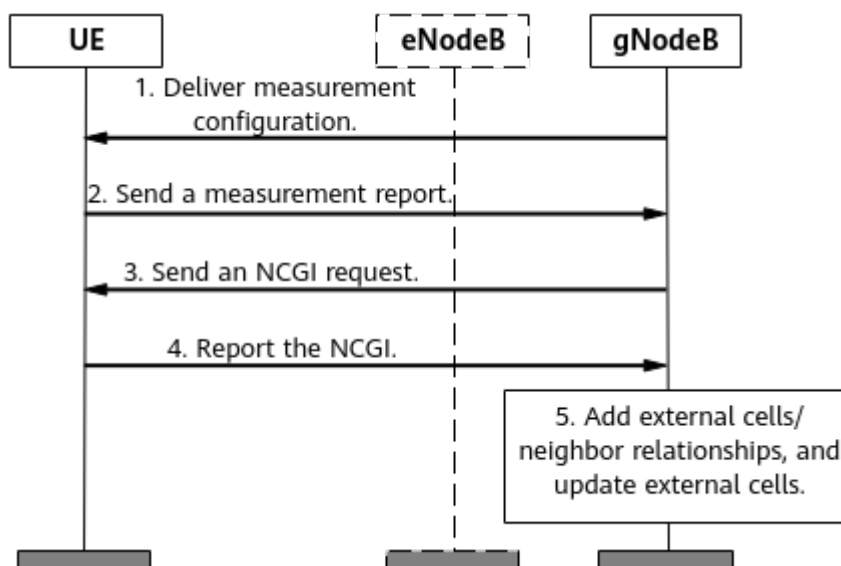
- Proactive PCI conflict detection can be triggered only based on intra-RAT ANR.
- Passive PCI conflict detection can be triggered based on intra-RAT and inter-RAT ANR.

Triggering of proactive PCI conflict detection based on ANR

After intra-RAT ANR is enabled, the gNodeB can proactively trigger PCI conflict detection based on intra-RAT ANR.

This function is controlled by the **INTRA_RAT_ANR_PCI_CONFL_DET_SW** option of the **NRCellAlgoSwitch.AnrSwitch** parameter. When this option is selected and the **PCI_COLLISION_DETECT_SW** or **PCI_CONFUSION_DETECT_SW** option of the **NRCellAlgoSwitch.AnrSwitch** parameter is selected, the gNodeB instructs UEs to perform NCGI reading of neighboring cells and adds unknown neighboring cells within the period of [**NRCellAnr.StartTime**, **NRCellAnr.StopTime**] according to the procedure shown in [Figure 4-2](#). In this way, PCI conflict detection is triggered. The dashed lines indicate the messages exchanged between the gNodeB and a UE through the eNodeB in NSA networking.

Figure 4-2 Procedure for triggering proactive PCI conflict detection based on intra-RAT ANR



1. The gNodeB delivers the periodic measurement configuration to the UE. Within the period of [**NRCellAnr.StartTime**, **NRCellAnr.StopTime**], the gNodeB delivers configurations for frequency measurements based on certain rules, and randomly selects UEs that meet certain requirements at an interval of 30s for proactive PCI conflict detection. The gNodeB selects a maximum of five UEs in the serving cell at a time.

For details about how the gNodeB selects frequencies to be delivered for measurements and selects UEs, see *ANR*.

The gNodeB instructs the UEs to measure the operating frequency of the serving cell and all configured intra-RAT neighboring frequencies. If the number of frequencies to be measured is limited and not all the frequencies can therefore be delivered for measurements, the gNodeB selects frequencies based on their priorities. For details about configuring the candidate frequency list based on frequency priorities, see *NSA Mobility Management* and *SA Mobility Management in Connected Mode*.

The maximum number of periodic measurements of each frequency is 100. The rules for calculating the number of periodic measurements are as follows:

- When proactive PCI conflict detection based on intra-RAT ANR is enabled or after it is enabled the **NRCCELLANR.StartTime** arrives, the gNodeB sets the numbers of periodic measurements of the operating frequency of the serving cell and intra-RAT neighboring frequencies to zero.
 - After a periodic measurement of a frequency is complete, the gNodeB increases the number of periodic measurements for this frequency by one.
 - If the neighbor relationship with a cell on a frequency is successfully added by ANR, the gNodeB sets the number of periodic measurements of the frequency to zero. If this neighbor relationship is added by running MML commands, the gNodeB does not set this number to zero.
 - The number of periodic measurements of a new intra-RAT neighboring frequency is counted from zero.
2. The UE sends a measurement report to the gNodeB. The UE performs the measurements for four times and sends measurement reports at an interval of 5120 ms.
 3. The gNodeB selects the neighboring cell with the best signal quality and whose RSRP is greater than or equal to **NRCCellAnr.IntraRatAnrRSRPThld** in the measurement report, and sends an NCGI request to the UE.
 4. The UE reads SIB1 of this neighboring cell to obtain CGI-Info (including the NCGI, TAC, PCI, PLMN list, and SSB frequency), and reports it to the gNodeB.
 5. The gNodeB obtains the NCGI, TAC, PCI, PLMN list, SSB frequency, and other information of the neighboring cell from CGI-Info in the measurement report from the UE. Then, the gNodeB operates according to whether the neighboring cell is served by the gNodeB itself and whether it is included in the NRT and NCL of the gNodeB, as described in [Table 4-1](#).

Table 4-1 Operation performed by the gNodeB

Whether a cell is a cell served by the local base station	In the NRT or Not	In the NCL or Not	Operation Performed by the gNodeB
Yes	No	Not involved	Adds the neighboring cell to the NRT and trigger PCI conflict detection.

Whether a cell is a cell served by the local base station	In the NRT or Not	In the NCL or Not	Operation Performed by the gNodeB
Yes	Yes	Not involved	No need to operate.
No	No	No	Adds the neighboring cell to the NRT and NCL and triggers PCI conflict detection.
No	No	Yes	Adds the neighboring cell to the NRT, updates the NCL as required, and triggers PCI conflict detection.
No	Yes	Yes	Updates the NCL as required. If the SSB frequency or PCI of an external cell in the NCL is updated, PCI conflict detection will be triggered.

Triggering of passive PCI conflict detection based on ANR

When the gNodeB uses intra-RAT ANR or inter-RAT ANR to automatically add, update, or remove intra-RAT external cells or neighbor relationships or external LTE cells or neighbor relationships with LTE cells, NCL or NRT changes associated with PCI conflicts occur. As a result, the gNodeB starts PCI conflict detection. For details about intra- and inter-RAT ANR, see *ANR*.

Triggering of PCI Conflict Detection Based on Xn Messages

gNodeBs exchange information using Xn messages and update external cells in the case of an Xn interface setup or gNodeB configuration update at either end of the Xn interface. As a result, the gNodeB starts PCI conflict detection. This triggering method applies only to SA networking.

- Xn setup between gNodeBs
The local gNodeB exchanges information with the peer gNodeB through the Xn setup procedure.
After receiving an NCGI, SSB frequency, PCI, and other information carried in the *Served Cell Information NR* IE in an XN SETUP REQUEST or XN SETUP RESPONSE message, the gNodeB searches its NCL for the external cell corresponding to the NCGI. If the SSB frequency and PCI of the external cell in the NCL are different from the SSB frequency and PCI carried in the IE, the gNodeB updates the NCL using the SSB frequency or PCI carried in the IE and in turn triggers PCI conflict detection.
- gNodeB configuration change
The local gNodeB exchanges information with the peer gNodeB through the NG-RAN node configuration update procedure.

After receiving an NCGI, SSB frequency, PCI, and other information carried in the *Served Cell Information NR* IE in an NG-RAN NODE CONFIGURATION UPDATE or NG-RAN NODE CONFIGURATION UPDATE ACKNOWLEDGE message, the gNodeB searches its NCL for the external cell corresponding to the NCGI. If the SSB frequency and PCI of the external cell in the NCL are different from the SSB frequency and PCI carried in the IE, the gNodeB updates the NCL using the SSB frequency or PCI carried in the IE and in turn triggers PCI conflict detection.

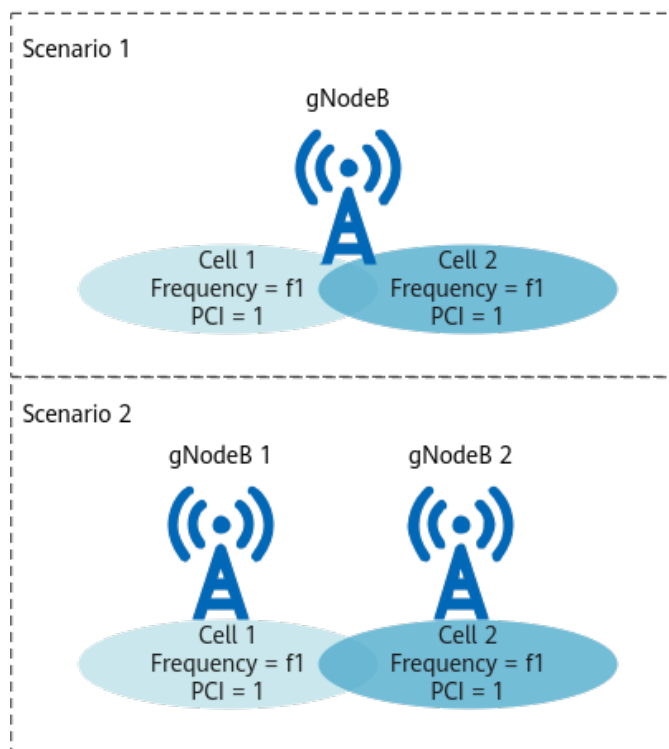
4.1.1.2 PCI Conflict Identification

After PCI conflict detection is triggered, the gNodeB identifies PCI collisions and PCI confusions separately.

PCI Collision

The gNodeB checks the NR cells that it serves to determine whether they work on the same frequency and use the same PCI. It also checks whether these NR cells work on the same frequency and use the same PCI as external NR cells in its intra-RAT NCL. If the gNodeB identifies such cells, it identifies the presence of PCI collisions. [Figure 4-3](#) illustrates an example where cell 1 and cell 2 work on the same frequency and use the same PCI. For details about PCI collision identification between neighboring LTE cells of an NR cell, see *PCI Conflict Detection and Self-Optimization* in *eRAN Feature Documentation*.

Figure 4-3 PCI collision identification



In scenario 1, the gNodeB detects that the NR cells it serves (cell 1 and cell 2) work on the same frequency and use the same PCI.

In scenario 2, two different gNodeBs serve the NR cells: Cell 1 is in the NCL of gNodeB 2, and cell 2 is in the NCL of gNodeB 1.

- gNodeB 1 detects that the NR cell it serves (cell 1) and an external NR cell (cell 2) work on the same frequency and use the same PCI.
- gNodeB 2 detects that the NR cell it serves (cell 2) and an external NR cell (cell 1) work on the same frequency and use the same PCI.

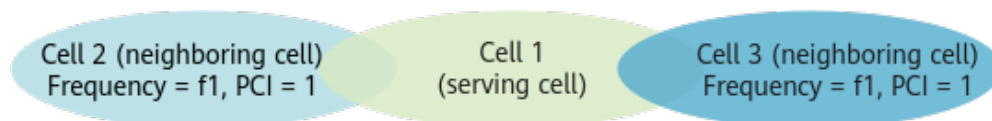
When the number of available PCIs is almost equal to the total number of NR cells on a frequency served by a gNodeB and external NR cells on the frequency in its NCL, PCI collisions cannot be resolved by PCI reallocation due to the limited number of PCIs. Therefore, when intra-frequency NR cells using the same PCI are not located within close proximity to each other (for example, in scenarios where RRUs are remotely installed), disabling PCI collision identification by deselecting the **PCI_COLLISION_DETECT_SW** option of the **NRCellAlgoSwitch.AnrSwitch** parameter is recommended.

PCI Confusion

When the NRT of a serving cell contains two intra-frequency neighboring cells using the same PCI (both are NR cells or LTE cells), the gNodeB can identify the presence of PCI confusions.

Figure 4-4 illustrates an example where cell 2 and cell 3 are intra-frequency NR cells (or LTE cells) using the same PCI and are both contained in the intra-RAT NRT (or inter-RAT NRT) of the serving NR cell (cell 1). In this case, the gNodeB identifies a PCI confusion between cell 2 and cell 3.

Figure 4-4 PCI confusion identification



4.1.1.3 PCI Conflict Reporting

After detecting a PCI conflict, the gNodeB generates ALM-29847 NR Cell PCI Conflict and reports the PCI conflict to the MAE-Access.

4.1.2 PCI Self-Optimization

The MAE-Access implements PCI self-optimization between NR cells following the method specified by the **Start optimization task** parameter.

- If **Start optimization task** is set to **Immediate**, the MAE-Access instantly implements PCI self-optimization between NR cells.
- If **Start optimization task** is set to **Periodic**, the MAE-Access implements PCI self-optimization between NR cells at a scheduled time every day.

Once the task is initiated, the MAE-Access performs PCI self-optimization between NR cells based on the PCI conflicts between NR cells reported by the gNodeB. Specifically, the MAE-Access:

1. Determines PCI reallocation priority. For details, see [4.1.2.1 Determination of PCI Reallocation Priorities](#).

2. Reallocates PCIs for conflicting cells to be optimized in descending order of PCI reallocation priority. For details, see [4.1.2.2 PCI Reallocation](#).
3. Delivers the reallocated PCIs to the gNodeB. For details, see [4.1.2.3 New PCI Delivery](#).

For details on PCI self-optimization between NR cells, refer to the related descriptions in *RAN Management* in *MAE Product Documentation (EulerOS, TaiShan)*.

For details about PCI self-optimization between LTE cells, see *PCI Conflict Detection and Self-Optimization* in *eRAN Feature Documentation*.

NOTE

PCI optimization cannot be performed across MAE-Access systems. An MAE-Access cannot optimize PCIs for neighboring cells that have PCI conflicts with its own cell if those neighboring cells are managed by a different MAE-Access system. Therefore, manually specifying PCIs for conflicting cells managed by different MAE-Access systems is recommended.

4.1.2.1 Determination of PCI Reallocation Priorities

If multiple cells have PCI conflicts, the MAE-Access reallocates PCIs to conflicting cells in descending order of PCI reallocation priority (1 > 2 > 3 > 4):

1. Cells with user-specified priorities

On the MAE-Access, **PCI optimization priority** can be set to **High**, **Low**, or **Locked** for an NR cell.

The MAE-Access first reallocates PCIs to conflicting cells whose **PCI optimization priority** is **High**, and then to conflicting cells whose **PCI optimization priority** is **Low**. The MAE-Access does not reallocate PCIs to conflicting cells whose **PCI optimization priority** is **Locked**.

2. Cells with high PCI conflict degrees

The PCI conflict degree reflects the number of cells that a cell has PCI conflicts with. A larger number of conflicting cells indicates a higher PCI conflict degree. PCI conflicts on the networks are significantly reduced by altering the PCI of a cell with a high conflict degree. When cells have the same **PCI optimization priority** setting, the MAE-Access preferentially reallocates PCIs to cells with a higher PCI conflict degree.

3. Cells with recently changed PCIs

Cells with recently changed PCIs (including newly deployed cells) may be more prone to PCI data errors than cells that have been successfully running for longer periods, and changing their PCIs has less of an impact on the live network. The MAE-Access determines PCI reallocation priority based on the parameters **Interval for defining old cells (days)** and **Interval for cells of same batch (days)**.

The MAE-Access checks if the number of days elapsed from a cell's last PCI change to the current day is greater than the value of **Interval for defining old cells (days)**.

- If it is, the MAE-Access determines that the cell is old.

All old cells are grouped into the same batch. These cells are not prioritized for PCI reallocation based on the time of their PCI change, and they have the same PCI reallocation priority.

- If it is not, the MAE-Access determines that the cell is new.
All new cells are divided into batches and prioritized for PCI reallocation as follows: Starting from the time when the last PCI change occurred, the MAE-Access separates time periods based on the value of **Interval for cells of same batch (days)**. Cells whose PCIs are changed in the same time period are grouped into the same batch. These cells are not prioritized for PCI reallocation based on the time of their PCI change, and they have the same PCI reallocation priority.

Cells in a cell batch with a more recent PCI change time have higher PCI reallocation priorities. When cells have the same **PCI optimization priority** setting and PCI conflict degree, the MAE-Access preferentially reallocates PCIs to cells in the batch with the most recent PCI change time.

4. Cells with few neighboring cells

During PCI reallocation, resetting a cell with fewer neighboring cells has less of an impact on the live network. When cells that have the same **PCI optimization priority** setting and PCI conflict degree are grouped in the same batch according to their PCI change time, the MAE-Access preferentially reallocates PCIs to cells with fewer neighboring cells.

In the following example, four cells (cell 1, cell 2, cell 3, and cell 4) use the same PCI, and twelve PCI conflict messages are reported to the MAE-Access. See [Figure 4-5](#).

Figure 4-5 PCI conflict example



If **PCI optimization priority** is set to **High** for cell 1 and **Low** for cell 2, cell 3, and cell 4, the MAE-Access preferentially reallocates a PCI to cell 1. After cell 1 obtains a new PCI, only six PCI conflict messages are reported to the MAE-Access. See [Figure 4-6](#).

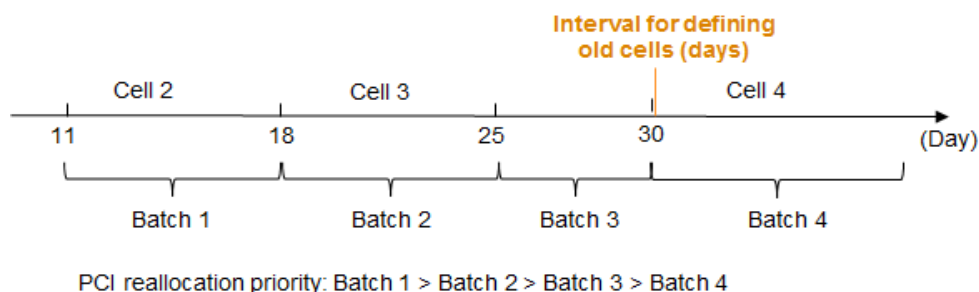
Figure 4-6 PCI conflict example



The MAE-Access checks the PCI conflict degree of cell 2, cell 3, and cell 4 because they have the same user-specified priority. As any of these cells has PCI conflicts with the other two cells, these cells have the same conflict degree. The MAE-Access then checks each of their PCI change times to determine the PCI reallocation priorities. If **Interval for defining old cells (days)** is set to 30, **Interval for cells of same batch (days)** is set to 7, and the number of days elapsed from the last PCI change for cell 2, cell 3, and cell 4 is 11, 19, and 40, respectively. As such, cell 4 is an old cell, while both cell 2 and cell 3 are new cells, as discussed in PCI reallocation priority 3 (cells with recently changed PCIs).

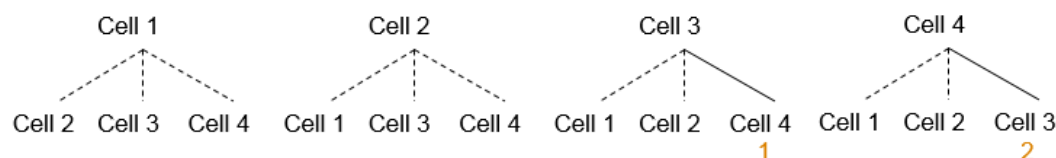
Therefore, the PCI reallocation priorities of the three cells are as follows: cell 2 > cell 3 > cell 4. See [Figure 4-7](#).

Figure 4-7 Determining the PCI reallocation priority



According to the determined priorities, the MAE-Access preferentially reallocates a PCI to cell 2. After cell 2 obtains a new PCI, only two PCI conflict messages are reported to the MAE-Access. See [Figure 4-8](#).

Figure 4-8 PCI conflict example



The MAE-Access preferentially reallocates a PCI to cell 3 because cell 3 has a higher PCI reallocation priority than cell 4. After cell 3 obtains the new PCI, no PCI conflict message is reported to the MAE-Access.

4.1.2.2 PCI Reallocation

The MAE-Access reallocates PCIs to conflicting cells to be optimized after PCI reallocation priority is determined. PCIs from 0 to 1007 can be used for NR cells by default. **NR-NR PCI Range** can be configured if conflicting cells need PCIs to be reallocated to them within a specific PCI range. PCI reallocation is implemented according to the following principles:

- Each new PCI allocated to a conflicting cell cannot collide with those of other cells. For details on PCI collision, see [PCI Collision](#).

When the **PCI_COLLISION_DETECT_SW** option of the **NRCCellAlgoSwitch.AnrrSwitch** parameter is selected, it is recommended that **PCI reallocation optimization** be turned off on the MAE-Access. In this case, the following principles for PCI reallocation also apply:

- Each new PCI allocated to a conflicting cell cannot be the same as that of any intra-frequency external cell in the NCL of the gNodeB that serves the conflicting cell.
- Each new PCI allocated to a conflicting cell cannot be the same as that of any intra-frequency cell served by the same gNodeB as the conflicting cell.

When the **PCI_COLLISION_DETECT_SW** option of the **NRCCellAlgoSwitch.AnrrSwitch** parameter is deselected, it is recommended

that **PCI reallocation optimization** be turned on on the MAE-Access. In this case, the principles for PCI reallocation do not apply.

- Each new PCI allocated to a conflicting cell cannot confuse with those of other cells. For details on PCI confusion, see [PCI Confusion](#).
- When PCI-related parameters are all configured correctly, the following principles for PCI reallocation also apply:
 - In PCI reallocation, try to ensure that PCI mod 30 returns different values for a conflicting cell and its adjacent cells.

When PCI mod 30 returns the same value for different cells, the uplink reference signal sequence group numbers are also the same for these cells. As such, their uplink signals interfere with each other. For details, see section 6.4.1 "Reference signals" in 3GPP TS 38.211 V15.5.0.
 - In PCI reallocation, try to ensure that (PCI plus offset) mod 30 returns different values for a conflicting cell and its adjacent cells.

This principle further prevents interference between uplink reference signals of cells with the same uplink reference signal sequence group number. The value of (PCI plus offset) mod 30 for the conflicting cell is allocated by the MAE-Access. The offset is automatically calculated by the MAE-Access.
 - In PCI reallocation, try to ensure that PCI mod 3 returns different values for a conflicting cell and its adjacent cells.

When PCI mod 3 returns different values for different cells, primary synchronization code sequences are also different for these cells. For details, see section 7.3.2.2 "Primary synchronization signal" in 3GPP TS 38.211 V15.5.0.
 - In PCI reallocation, try to ensure that PCI mod 4 returns different values for a conflicting cell and its adjacent cells.

When PCI mod 4 returns different values for different cells, Physical Broadcast Channel (PBCH) Demodulation Reference Signal (DMRS) frequency-domain positions are staggered for these cells. For details, see section 7.4.3 "SS/PBCH block" in 3GPP TS 38.211 V15.5.0.
 - If the new PCI to be allocated to a conflicting cell needs to be used by another cell, ensure that the distance between the two cells is enough for them to be served by different gNodeBs which are as far apart as possible.

If some of the engineering parameters are not configured or configured incorrectly, the new PCI allocated to a conflicting cell may not be the optimal one, adversely affecting PCI optimization. This is because the new PCI's ability to solve PCI conflicts is the only factor considered during PCI self-optimization, as opposed to the different returned values of PCI mod 30, PCI mod 3, and PCI mod 4 for the conflicting cell and its adjacent cells.

4.1.2.3 New PCI Delivery

The MAE-Access delivers new PCIs to the gNodeB in a method specified by the **Apply optimization advice** parameter.

- When **Apply optimization advice** is set to **Immediate**, the MAE-Access delivers the new PCIs to the gNodeB immediately after PCI self-optimization is completed.

- When **Apply optimization advice** is set to **Scheduled**, the MAE-Access delivers the new PCIs to the gNodeB at a scheduled time every day.
- When **Apply optimization advice** is set to **Manual**, the new PCIs need to be manually confirmed before being delivered to the gNodeB.

NOTICE

Before delivering new PCIs in NSA networking, ensure that the **MOD_NCELL_CFG_SW** option of the **GlobalProcSwitch.X2BasedUptNcellCfgSwitch** parameter has been selected on the LTE side.

4.2 Network Analysis

4.2.1 Benefits

It is recommended that the PCI conflict detection and self-optimization function be always enabled on commercial networks to help promptly detect and optimize any potential PCI conflicts. When the total number of intra-frequency cells served by a gNodeB and intra-frequency external cells in the NCL of the gNodeB approximates the number of available PCIs, PCIs will be reused (indicating that the same PCI is used by multiple cells). In practice, cells with the same PCI but are far away from each other (for example, in scenarios where RRUs are remotely installed) do not interfere with each other. In such a case, it is recommended that only PCI confusion detection be enabled and PCI collision detection be disabled to prevent incorrect PCI collision detection.

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

PCI conflict detection consumes significant gNodeB system resources (including CPU and memory resources). Delivering new PCIs during PCI self-optimization causes cells to restart automatically, which interrupts ongoing services carried on such cells and affects UE services.

4.3 Requirements

4.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-021204	Automatic Neighbour Relation (ANR)	NR0S0AUTNE00	per Cell

NOTE

If this license has not been activated, the gNodeB will not report PCI conflicts to the MAE-Access, and PCI conflict information will not be displayed in the **PCI Conflict Optimization** window or in the SON logs.

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	NR PCI conflict self-optimization	None	<i>RAN Management</i> in <i>MAE Product Documentation</i> (<i>EulerOS, TaiShan</i>)	NR PCI conflict optimization in the OSS feature WOFD-905207 Network Planning & Optimization Expert - NR must be enabled prior to PCI conflict detection and self-optimization.
Low-frequency TDD	Intra-RAT ANR	NR_NR_ANR_SW option of the NRCellAlgoSwitch . AnrSwitch parameter	<i>ANR</i>	Intra-RAT ANR must be enabled before PCI conflict detection based on intra-RAT ANR or Xn messages is enabled.
Low-frequency TDD	Inter-RAT ANR	NR_EUTRAN_ANR_SW option of the NRCellAlgoSwitch . AnrSwitch parameter	<i>ANR</i>	Inter-RAT ANR must be enabled before PCI conflict detection based on inter-RAT ANR is enabled.

4.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PCI-specific intra-RAT ANR	PCI_SPECIFIC_NR_A_NR_SW option of the NRCellAlgoSwitch . AnrSwitch parameter	<i>ANR</i>	Proactive PCI conflict detection based on intra-RAT ANR is mutually exclusive with PCI-specific intra-RAT ANR.
Low-frequency TDD	Intra-RAT fast ANR	NR_NR_FAST_ANR_SW option of the NRCellAlgoSwitch . AnrSwitch parameter	<i>ANR</i>	Proactive PCI conflict detection based on intra-RAT ANR is mutually exclusive with intra-RAT fast ANR.

4.3.2.3 Function Impacts

None

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

This function does not depend on RF modules.

4.3.4 Others

PCI self-optimization is supported in MAE V100R020C10 and later.

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

PCI Conflict Detection

[Table 4-2](#) describes the parameters used for activating PCI conflict detection, and [Table 4-3](#) describes optimization parameters. This section does not describe parameters related to cell establishment.

Table 4-2 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
ANR Switch	NRCellAlgoSwitch . <i>AnrSwitch</i>	PCI_COLLISION_DETECT_SW	This option specifies whether to enable PCI collision detection between NR cells. It is recommended that this option be selected except when RRUs are remotely installed.
ANR Switch	NRCellAlgoSwitch . <i>AnrSwitch</i>	PCI_CONFUSION_DETECT_SW	This option specifies whether to enable PCI confusion detection between NR cells. It is recommended that this option be selected.
ANR Switch	NRCellAlgoSwitch . <i>AnrSwitch</i>	EUTRAN_PCI_CONFUSION_DETECT_SW	This option specifies whether to enable PCI confusion detection between neighboring LTE cells of an NR cell. It is recommended that this option be selected.

Parameter Name	Parameter ID	Option	Setting Notes
ANR Switch	NRCellAlgoSwitch . <i>AnrSwitch</i>	INTRA_RAT_ANR_PCI_CONFL_DET_SW	This option specifies whether to enable proactive PCI conflict detection based on intra-RAT ANR. It is recommended that this option be selected to enable proactive PCI conflict detection based on intra-RAT ANR.

Table 4-3 Parameters used for optimization

Parameter Name	Parameter ID	Setting Notes
Start Time	NRCellAnr . <i>StartTime</i>	Retain the default value.
End Time	NRCellAnr . <i>StopTime</i>	Retain the default value.
Intra-RAT ANR RSRP Threshold	NRCellAnr . <i>IntraRatAnrRSRPTHld</i>	Retain the default value.

PCI Self-Optimization

Table 4-4 describes the parameters used for activating PCI self-optimization. No optimization parameters are involved.

Table 4-4 Parameters used for activation (MAE-Access)

Parameter Name	Parameter ID	Setting Notes
Interval for defining old cells (days)	None	The recommended value is 30 .
Interval for cells of same batch (days)	None	The recommended value is 7 . The value of Interval for cells of same batch (days) cannot be greater than the value of Interval for defining old cells (days) .

Parameter Name	Parameter ID	Setting Notes
PCI range	None	This parameter can be set to a consecutive or discrete PCI range. Set this parameter for MAE-Access border cells based on PCI ranges configured on the peer MAE-Access systems.
PCI optimization priority	None	The default value is Low . When this parameter is set to Locked , PCI self-optimization can easily fail. Therefore, exercise caution when setting this parameter.
PCI reallocation optimization	None	The recommended value is Off when the PCI_COLLISION_DETECT_SW option of the NRCellAlgoSwitch.AnrSwitch parameter is selected. In other cases, the recommended value is On .
Start optimization task	None	- Set this parameter to Immediate to immediately start PCI self-optimization. - Set this parameter to Periodic to enable scheduled PCI self-optimization.
Apply optimization advice	None	- Set this parameter to Manual if new PCIs need to be manually confirmed before being delivered. - Set this parameter to Immediate or Scheduled as required if new PCIs do not need to be manually confirmed before being delivered.

4.4.1.2 Using MML Commands

4.4.1.2.1 PCI Conflict Detection

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

- PCI collision detection between NR cells and PCI confusion detection between NR cells

```
//Enabling PCI collision detection between NR cells
MOD NRCELLALGOSWITCH: NrCellId=0, AnrSwitch=PCI_COLLISION_DETECT_SW-1;
//Enabling PCI confusion detection between NR cells
MOD NRCELLALGOSWITCH: NrCellId=0, AnrSwitch=PCI_CONFUSION_DETECT_SW-1;
//Enabling proactive PCI conflict detection based on ANR
MOD NRCELLALGOSWITCH: NrCellId=0, AnrSwitch=INTRA_RAT_ANR_PCI_CONFL_DET_SW-1;
```

- PCI confusion detection between neighboring LTE cells of an NR cell
 - a. On the MAE-Access, choose **SON > NR PCI Conflict Optimization > NR-LTE PCI Conflict**. On the displayed tab page, turn on **NR-LTE PCI Conflict Switch**.
 - b. Run the following MML commands:
MOD NRCELLALGOSWITCH: NrCellId=0, AnrSwitch=EUTRAN_PCI_CONFUSION_DETECT_SW-1;

Optimization Command Examples

```
//Optimizing the parameters for proactive PCI conflict detection based on ANR
MOD NRCELLANR: NrCellId=0, StartTime=14&00&00, StopTime=15&00&00, IntraRatAnrRsrpThld=-102;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- PCI collision detection between NR cells and PCI confusion detection between NR cells
//Disabling proactive PCI conflict detection based on ANR
MOD NRCELLALGOSWITCH: NrCellId=0, AnrSwitch=INTRA_RAT_ANR_PCI_CONFL_DET_SW-0;
//Disabling PCI collision detection between NR cells
MOD NRCELLALGOSWITCH: NrCellId=0, AnrSwitch=PCI_COLLISION_DETECT_SW-0;
//Disabling PCI confusion detection between NR cells
MOD NRCELLALGOSWITCH: NrCellId=0, AnrSwitch=PCI_CONFUSION_DETECT_SW-0;
- PCI confusion detection between neighboring LTE cells of an NR cell
 - a. Run the following MML commands:
MOD NRCELLALGOSWITCH: NrCellId=0, AnrSwitch=EUTRAN_PCI_CONFUSION_DETECT_SW-0;
 - b. On the MAE-Access, choose **SON > NR PCI Conflict Optimization > NR-LTE PCI Conflict**. On the displayed tab page, turn off **NR-LTE PCI Conflict Switch**.

4.4.1.2.2 PCI Self-Optimization

Prerequisites

- Configurations of available PCI ranges, PCI optimization priorities, and PCI reallocation optimization
Configure them directly on the MAE-Access or by importing the engineering parameter template into the MAE-Access. To import the engineering parameter template into the MAE-Access, prepare the template as shown in [Figure 4-9](#).

Figure 4-9 Engineering parameter template example

A	B	C	D	E	F	G
MCC	MNC	gNodeB ID	Cell ID	Range	PCI Optimization Priority	PCI Reallocation Optimization

- Engineering parameter configurations related to PCI self-optimization
Configure them using MML commands or by importing the engineering parameter template into the MAE-Access. To import the engineering

parameter template into the MAE-Access, prepare the template as shown in [Figure 4-10](#).

Figure 4-10 Engineering parameter template example

A	B	C	D	E	F	G	H	I	J	K
MCC	MNC	gNodeBId	CellId	SectorId	Longitude	Latitude	Azimuth	Beamwidth	DeviceNo	SubunitNo

If the engineering parameters of a cell have been configured by running MML commands and importing the engineering parameter template into the MAE-Access, the longitude, latitude, and azimuth specified in the engineering parameter template on the MAE-Access as well as the beamwidth configured by running MML commands all take effect.

Activation

NOTICE

- The delivery time of new PCIs must be different from the execution time of a PCI self-optimization task. This ensures that PCIs are not delivered during PCI self-optimization.
- A short interval needs to be maintained between the completion of a PCI self-optimization task and the delivery of new PCIs. This ensures that PCIs are calculated based on up-to-date PCI conflict information. Networks are particularly prone to frequent parameter setting changes during their initial-phase deployment; therefore, delivering new PCIs immediately after PCI self-optimization is complete is recommended.
- It is recommended that PCIs be delivered in off-peak hours. This is because the cell will be automatically reset when the MAE-Access delivers PCI optimization suggestions.

To create and start a PCI self-optimization task, perform the following steps:

Step 1 On the MAE-Access, choose **SON > NR PCI Conflict Optimization**. The **NR PCI Conflict Optimization** page is displayed.

On the **PCI Conflict Information** page in the right pane, click **Refresh** to view the PCI conflict information reported by all NEs managed by the MAE-Access.

Step 2 (Optional) Configure PCI self-optimization parameters.

1. When the cell engineering parameters are updated, the parameters need to be re-imported.

In the left navigation pane, choose **NR-NR PCI Conflict > PCI Conflict Information**. On the **PCI Conflict Information** page in the right pane, click **Engineering Parameters** and choose **Import** from the drop-down menu to import the prepared cell engineering parameter file.

2. Configure the available PCI range, PCI optimization priority, and PCI reallocation optimization function for a cell as required.

In the left navigation pane, choose **NR-NR PCI Range**. On the **NR-NR PCI Range** page in the right pane, click **Import** to import the prepared

engineering parameter template. Alternatively, click **Setting** to configure the available PCI range, PCI optimization priority, and PCI reallocation optimization switch for the cell.

 NOTE

If some of the engineering parameters are not configured or configured incorrectly, the new PCI allocated to a conflicting cell may not be the optimal one, adversely affecting PCI optimization. This is because the new PCI's ability to solve PCI conflicts is the only factor considered during PCI self-optimization, as opposed to the different returned values of PCI mod 30, PCI mod 3, and PCI mod 4 for the conflicting cell and its adjacent cells.

Step 3 Start a PCI self-optimization task.

In the left navigation pane, choose **NR-NR PCI Conflict > Optimization Task**. On the **Optimization Task** page in the right pane:

1. (Optional) Click **Setting** and configure **Interval for defining old cells (days)** and **Interval for cells of same batch (days)**.
2. Click **Launch** and configure **Start optimization task** and **Apply optimization advice**.

----End

Optimization

N/A

Deactivation

After PCI conflict detection is disabled, PCI self-optimization becomes invalid. For details on how to deactivate PCI conflict detection, see [Deactivation Command Examples](#).

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

PCI Conflict Detection

Use any of the following three methods to verify that PCI conflict detection has taken effect.

- Viewing PCI conflict alarms
Check if ALM-29847 NR Cell PCI Conflict is reported to the MAE-Access. If it is, this function has taken effect.
- Viewing PCI conflict information
 - On the MAE-Access, choose **SON > NR PCI Conflict Optimization > NR-NR PCI Conflict**. Check if NR-NR PCI conflict information is displayed in

the **PCI Conflict Info** area. If it is displayed, PCI collision detection between NR cells and PCI confusion detection between NR cells have taken effect.

- On the MAE-Access, choose **SON > NR PCI Conflict Optimization > NR-LTE PCI Conflict**. Check if NR-LTE PCI conflict information is present. If it is displayed as shown in [Figure 4-11](#), PCI confusion detection between neighboring LTE cells of an NR cell has taken effect.

Figure 4-11 Information about conflicts between neighboring LTE cells of an NR cell

PCI Conflict Info						PCI Conflict Neighboring Cell
Cell CGI (MCC-MNC-gNodeBId-CellId)	Cell Name	NE Name	Conflict SSB Freq	Conflict Cell PCI	Conflict Generated On	Cell CGI (MCC-MNC-eNodeBId-CellId)
460-10-478882-15	cell_15	cn_0830	500	2	2021-02-03 17:14:23 +08:00	460-10-202020-3
						460-10-202020-4
460-10-478882-15	cell_15	cn_0830	500	3	2021-02-03 17:14:23 +08:00	460-10-202020-5
						460-10-202020-6
460-10-478882-15	cell_15	cn_0830	500	5	2021-02-03 17:14:23 +08:00	460-10-202020-9
						460-10-202020-10
460-10-478882-15	cell_15	cn_0830	500	6	2021-02-03 17:14:23 +08:00	460-10-202020-7
						460-10-202020-11

- Viewing SON logs

On the MAE-Access, choose **SON > SON Log**. Click **Query SON Log** in the navigation pane, and then click **Query** in the displayed **Query SON Log** tab page. Select **NR PCI Conflict Detection and Optimization Log** for **Log Category** and then click **OK**. If SON logs containing PCI conflict information are displayed, the function has taken effect.

PCI Self-Optimization

Use either of the following two methods to verify that PCI self-optimization has taken effect.

- Viewing PCI optimization suggestions

On the MAE-Access, choose **SON > NR PCI Conflict Optimization > NR-NR PCI Conflict**. On the displayed tab page, if new PCIs are displayed under **Recommended PCI** in the **PCI Conflict Neighboring Cell** area and current PCIs and recommended PCIs are displayed in the **Optimization Advice** area, PCI self-optimization has taken effect.

- Viewing SON logs

On the MAE-Access, choose **SON > SON Log**. Click **Query SON Log** in the navigation pane, and then click **Query** in the displayed **Query SON Log** tab page. Select **NR PCI Conflict Detection and Optimization Log** for **Log Category** and then click **OK**. If SON logs containing PCI self-optimization information are displayed, the function has taken effect.

4.4.3 Network Monitoring

None

5 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

6 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

7 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

8 Reference Documents

- *ANR*
- *NSA Mobility Management*
- *SA Mobility Management in Connected Mode*
- *RAN Management in MAE Product Documentation (EulerOS, TaiShan)*
- *PCI Conflict Detection and Self-Optimization in eRAN Feature Documentation.*
- *Technical Specifications in 3900 & 5900 Series Base Station Product Documentation*
- 3GPP TS 38.211: "NR; Physical channels and modulation"
- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"